

CONNECTED CAMERAS

*COMPETITIVE PRODUCT EXPERIENCE ANALYSIS
FROM AN AMATEUR PHOTOGRAPHER PERSPECTIVE*

STA Product & Experience Planning

November 16, 2012

Competitive Shootout

A good **connected camera** enables a photographer to capture and share planned or unplanned **precious moments** in an “**instant**” with minimal effort and the best quality possible.

Connected Cameras



Samsung WB850F
(similar sensor /optics as Galaxy Camera)



Apple iPhone5
(smartphone known for camera quality)



Nikon S800c
(first Digital Camera with Android OS in market)



Galaxy Camera
(first Android Camera with 4G connectivity)

Samsung WB850F Physical Fitness

From Pocket to Capture



[Watch video](#)

Samsung WB850F Physical Fitness

Borderline Pocketable: The barrel lens may get stuck in pocket at times but it is not uncomfortable to draw it and get ready to shoot pictures.

A bit larger than competitive DSCs.



Samsung WB850F Camera Features

Complex, Capable
with
Split Personalities

[Watch video](#)

Samsung WB850F Camera Features (Wi-Fi/GPS)

Attractive visual UI with help



Scene mode 1
physical mode dial



Program Mode
With Help

Very Rich but complex features



Dazzling Special
Effects



PASM
(Program, Aperture
Shutter, Manual)

Disparate UI paradigms



Scene mode 2
menu button



Scene mode 3
Fn button

Hard to discover capabilities



Smart filters only
available/visible in
PASM and Movie



Using Manual
Focus in this
camera? Wow..

Everything under the Kitchen Sink, overwhelming for casual users.

Samsung WB850F Camera Performance

Auto mode is usually good enough



Outstanding
candle light pictures

f/3.2 1/45s ISO240 8mm
(35mm equivalent: 42mm)

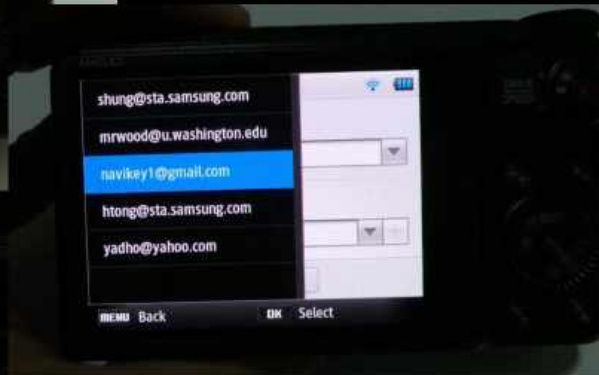


Great outdoor
pictures

f/5.8 1/500s ISO100 73mm
(35mm equivalent: 411mm)

Good picture quality with typical usage.

Samsung WB850F Connectivity & Sharing



Thoughtful design of Wi-Fi, Connectivity
& Sharing experience

Remote Viewfinder easy
to connect

Smooth connectivity and sharing experience without touch.

Samsung WB850F Battery Fitness

Initial Usage Experience:
Leaking Battery leaving a
bad taste...

SHOP ALL DEPARTMENTS • SEARCH: What do you want to find?   CART (0)

Home • Electronics • Cameras & Camcorders • Point & Shoot Cameras • Compare up to 4 products

16.2 MEGAPIXELS
3.0" AMOLED
21X ZOOM



\$100 OFF

Samsung WB850F WiFi Smart Digital Camera Bundle

Compare products:
Rated ★★★★★ (9 out of 73 reviews)
Share this Product:    

Item # 90850

Features:
Samsung Care, 8GB SDHC Card, Instant Photo Sharing, 1080p HD Movie Recording, Remote Viewfinder, Smartphone/Tablet Link

Product Video 

\$100 manufacturer's discount is valid 10/1/12 through 11/30/12. While supplies last.**

Online Price **\$349.99**
Less **-\$100.00**
Your Price **\$249.99**

Shipping & Handling included*

QUANTITY 1

Add to Cart

The estimated delivery time will be approximately 3 - 5 business days from the time of order.

PEOPLE WHO VIEWED

16.2 MEGAPIXELS
3.0" AMOLED
30x zoom



Fujifilm Finepix HS30EXR Digital Camera Bundle
★★★★★ (2)
~~\$369.99~~

16.2 MEGAPIXELS
3.0" AMOLED
18x zoom



Nikon Coolpix S9300 Digital Camera Bundle
★★★★★ (31)

Print this Page

Product Details Specifications Shipping & Terms Return Policy Free Concierge Service Reviews (20)

Firmware Update Available:
Updated firmware for this model is available that addresses a battery drain issue affecting some units. Please visit <http://www.samsung.com/us/support/owners/product/EC-WB850FBPBUS> to see Samsung's support webpage for the WB850F. The latest firmware can be found by clicking on "Manuals & Downloads" and then "Firmware".

Brilliant Photos. Instant Sharing. Now That's Smart.
The WB850F Samsung SMART Camera allows you to zoom in close, capture stunning photos using a 16MP BSI CMOS Sensor, and share them instantly with family and friends. Using the camera's built-in wireless functionality, you can upload high quality photos to social networking sites and seamlessly sync your camera to other SMART devices (TV, phone, tablet).

Specifications:

- **Effective Pixels:** 16.2 MP 1/2.3" BSI CMOS Sensor
- **Lens Specifications:** 21x Optical zoom lens (23-483mm) 4x Digital zoom (84x total zoom)
- **LCD Monitor:** 3.0" AMOLED (921,000 dot)
- **Motion Video Quality:** 1080p Full HD Video at 30fps
- **Image Stabilization:** Dual Image Stabilization
- **ISO Sensitivity:** Auto, ISO 100, 200, 400, 800, 1600, 3200
- **Images Per Battery Charge:** Approx. 200 shots
- **Dimensions:** 4.31"W x 2.44"H x 0.98"D (without protrusions)
- **Weight:** Approx. 8.0 oz
- **Storage Media:** SD / SDHC / SDXC

Features:

- **Your Best Photos Instantly Shared** - Instantly send quality photos to a small group of people directly from the camera. As connected devices with Samsung's TV Link and MobileLink, directly from the camera to Microsoft SkyDrive.
- **Remote Viewfinder** - Operates your camera directly from you
- **Smartphone Link** - Store and access your camera photos on
- **Easier to Take, Store and Share Pictures** - Use Auto Backup with the touch of a button. (also, enjoy the ultimate picture

Product Details Specifications Shipping & Terms Return Policy Free Concierge Service Reviews (20)

Firmware Update Available:

Updated firmware for this model is available that addresses a battery drain issue affecting some units. Please visit <http://www.samsung.com/us/support/owners/product/EC-WB850FBPBUS> to see Samsung's support webpage for the WB850F. The latest firmware can be found by clicking on "Manuals & Downloads" and then "Firmware".

Brilliant Photos. Instant Sharing. Now That's Smart.

The WB850F Samsung SMART Camera allows you to zoom in close, capture stunning photos using a 16MP BSI CMOS Sensor, and share them instantly with family and friends. Using the camera's built-in wireless functionality, you can upload high quality photos to social networking sites and seamlessly sync your camera to other SMART devices (TV, phone, tablet).

Battery issues resolved by firmware update..

Samsung WB850F: Lessons Learned

Nikon CoolPix 5100 Bridge Camera surpasses WB850F in both performance and image quality. Autofocus capability of WB850F is lagging behind in Auto Mode.

Nikon CoolPix 5100



f/5.1 1/500s ISO100 134mm
(35mm equivalent: 750mm)

The Samsung WB850F auto focus did not work well when zooming in and could not lock on the dragonfly. Note that max optical zoom (41X) was used to capture the dragonfly with Nikon 5100.

Samsung WB850F



f/4.8 1/1000s ISO100 39mm
(35mm equivalent: 217mm)

Galaxy Camera has autofocus problems under more extreme situations like hunting focus in scenes with variegated background and moving subjects (especially in video recording). Nikon s800c is also plagued with auto focus issues. Among the 4 devices tested, Apple iphone5 auto focus seems to be most versatile (but it has no optical zoom).

Not for non-casual photographers as primary camera.

iPhone5 Physical Fitness

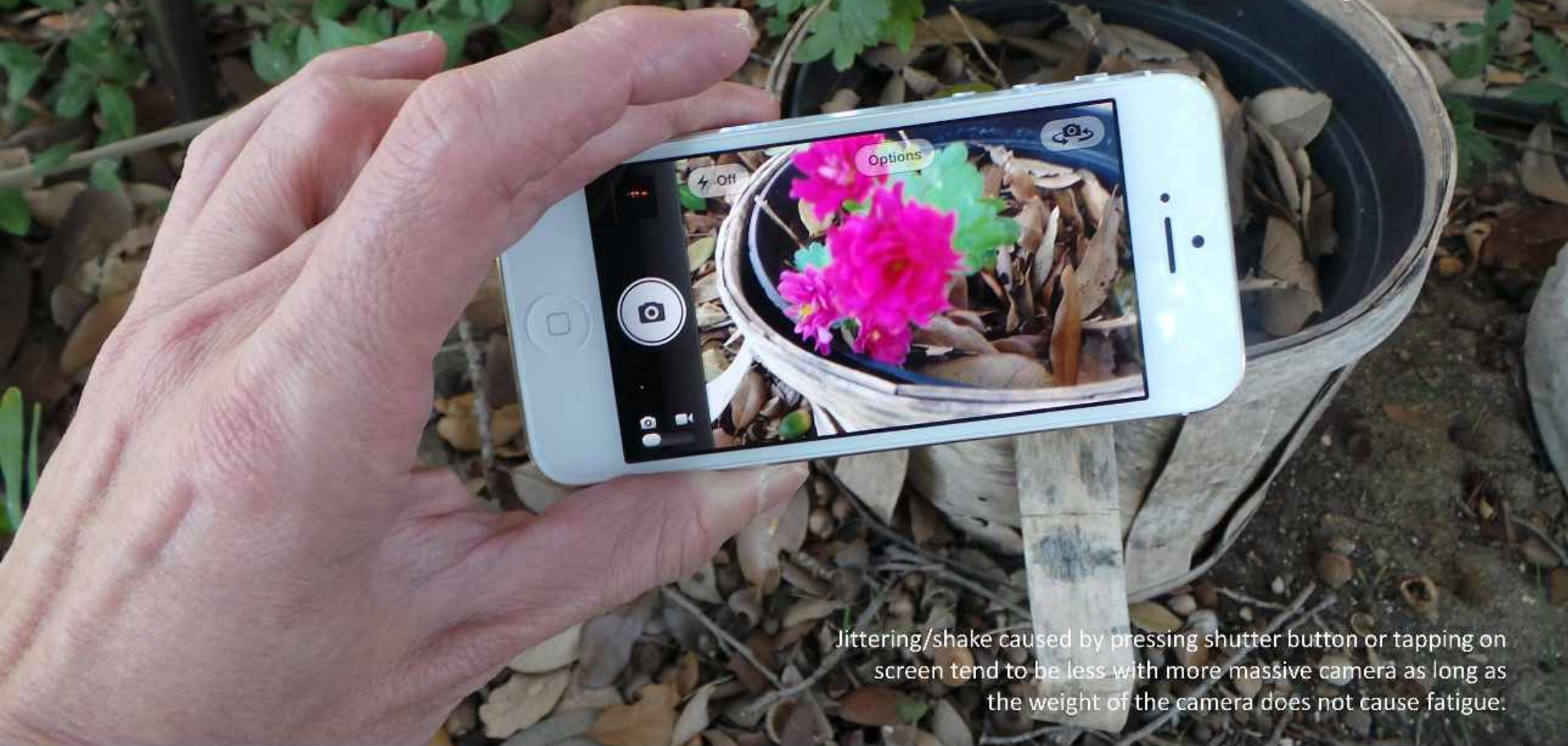
Instant Capture of
Precious Moment



[Watch video](#)

iPhone5 Physical Fitness

The quickest to draw from pocket, launch camera and take a picture.



Jittering/shake caused by pressing shutter button or tapping on screen tend to be less with more massive camera as long as the weight of the camera does not cause fatigue.

Quickest to take pictures but its light weight may amplify shake.

iPhone5 Physical Fitness



Location of camera lens is annoying. It was not designed for landscape orientation.

Finger-tainted pictures...

One has to be conscientious about holding the device in a way that any finger will not cover the lens. For some photographers, this may also mean a position more prone to shaking or movement of the camera.



Finger-unfriendly camera lens location in landscape orientation.

iPhone5 Camera & Photography Ecosystem



Default Camera has **barebone** features: Grid, HDR, Panorama and Flash

But with great performance & numerous third-party camera apps in AppStore

Plus a staggering number of third-party camera accessories (e.g., amazon.com)

http://www.amazon.com/s/ref=nb_sb_ss_l_2_14?url=search-alias%3Daps&field-keywords=iphone+camera+accessories&prefix=iphone+camera+%2Caps%2C10



Camera+



12x telephoto lens with tripod



LED light for iPhone camera

Interesting Camera App Demo Video

[Watch Video](#)



Camera!

Default camera performance is robust in many extreme situations.

iPhone5 Camera Performance

Image quality is comparable, if not better than many mid-range digital cameras, even in a variety of challenging situations.

Versatile in low light



f/2.4 1/15s ISO3200 4mm
(35mm equivalent: 33mm)



f/2.4 1/16s ISO800 4mm
(35mm equivalent: 33mm)

clipped from a Samsung Swiss-Gear Backpack Photo (uncompressed)



f/2.4 1/15s ISO800 4mm
(35mm equivalent: 33mm)

Intelligent Camera with minimal features for different scenes

iPhone5 : Lessons Learned

Both Galaxy S3 and iPhone5 are quick to launch, but Samsung cameras require more user attention, know-how and time to operate for good pictures.

Apple iPhone 5
Default Camera



f/2.4 1/15s ISO3200 4mm
(35mm equivalent: 33mm)

Samsung Galaxy S3
Single Shot Shooting Mode



f/2.6 1/15s ISO800 4mm
(35mm equivalent: 33mm)

Samsung Galaxy S3
Low Light Shooting Mode



Parameters unknown (not
reported by camera)

Need: Put more intelligence in our cameras, not more options!

* Intelligence = Imaging Domain Knowledge to take picture automatically

Nikon S800c Physical Fitness *(Ergonomics)*

Call to Action



[Watch video](#)

Nikon S800c Physical Fitness *(Ergonomics)*



Ready to Shoot? Draw your camera! It is smooth and quick to draw without being caught inside the pocket.

Comfortable to hold with 2 hands, but not as comfortable with one-hand operation as Galaxy Camera.

The design does not support right hand grip well.



One-hand operation not as good as anticipated.

Nikon S800c Capture Fitness

Touchy Touch Shutter

[Watch video](#)

Nikon S800c Capture Fitness



Touch shutter is on by default and the active shutter area occupies almost $\frac{3}{4}$ of the touch screen, making it accident prone. The setting can be changed through the Android "Menu" key.

Touch shutter will be automatically enabled in some shooting modes even when AE/AF was selected before. Some modes did not save while other modes would save the previous settings. Touch Shutter settings depend on the shooting mode selected and can change completely with different shooting modes.



It is difficult to anticipate touch shutter (or AF/AE) behavior.

Nikon S800c Capture Fitness *(Camera UX)*



A shooting mode melee.
Six different shooting modes with
no manual control.

Easy auto mode? Auto Mode?
Is "Auto" easy enough?



Memory-taxing: 3 screens of
modes to scroll through.

Smart portrait and another portrait
mode within Scene?

KISS - Keep It Simple! Stupid!

Nikon S800c Android Fitness *(Android UX)*



Poor Integration between Camera and Android. Picture Review and Gallery have very different UX.

Antiquated Android experience.

Unattractive graphical user interface design with eye-straining small fonts and icons



Ginger Bread Man with a so-and-so digital camera

Nikon S800c Camera Performance

Battery Woes

[Watch video](#)

Nikon S800c Battery unFitness



What's wrong with this camera?

The camera battery indicator is full but there is only 20% charge left!



One cannot use the camera when charging it!

One cannot charge it when it is connected to my PC as USB drive.

It's a brick when juicing up. I've never seen a camera like this.

Nikon S800c Sharing Fitness *(Connectivity & Sharing UX)*



Poor sharing and connectivity experience.



The preloaded “Upload to smart device” (on camera) and “connect to s800c” in ICS and Jelly Bean based smartphones do not work (downloaded from Play Store). The COOLPIX “Simple setup” does not work.



It does not preload with Nikon’s “my Picturetown”

Arrrh..Could not complete Setup!

Nikon S800c Camera Performance

Performed poorly when zooming in low light.



f/5.8 1/2s ISO400 45 mm
(35mm equivalent: 250mm)



f/5.8 1/2s ISO400 45 mm
(35mm equivalent: 250mm)

clipped from a Samsung Swiss-Gear Backpack Photo (uncompressed for comparing image quality).

Below was clipped from the picture at which the camera could focus.

Camera could not focus at the same distance as iPhone 5 and Galaxy Camera.



f/3.2 1/4s ISO400 4mm
(35mm equivalent: 33mm)



f/3.2 1/2s ISO400 4mm
(35mm equivalent: 33mm)

Picture quality of a < \$200 DSC retailed @ \$349.

Nikon S800c: Lessons Learned

A camera with a smart OS is not necessarily smart. Its shooting mode and touchy touch shutter make the shooting experience frustrating. Battery life is poor and the gauge in the camera is inaccurate.

Picture quality is inferior. Pictures taken with zoom in low light are dismal. There is not much of Nikon S800c a photographer can rely on.

Nikon S800c (failed to focus)



f/5.8 1/2s ISO400 45 mm
(35mm equivalent: 250mm)

Samsung Galaxy Camera



f/5.9 1/8s ISO800 86mm
(35mm equivalent: 483mm)

Galaxy Camera Physical unFitness (*Ergonomics*)

Pocket Workout



[Watch video](#)

Galaxy Camera Physical Fitness *(Ergonomics)*

Gee! What a workout!
I could have torn my jean's pocket and
broken the flash!
With this size, I expect the huge barrel
lens is interchangeable.



One-hand maneuverability of shutter and zoom is comfortable enough though.

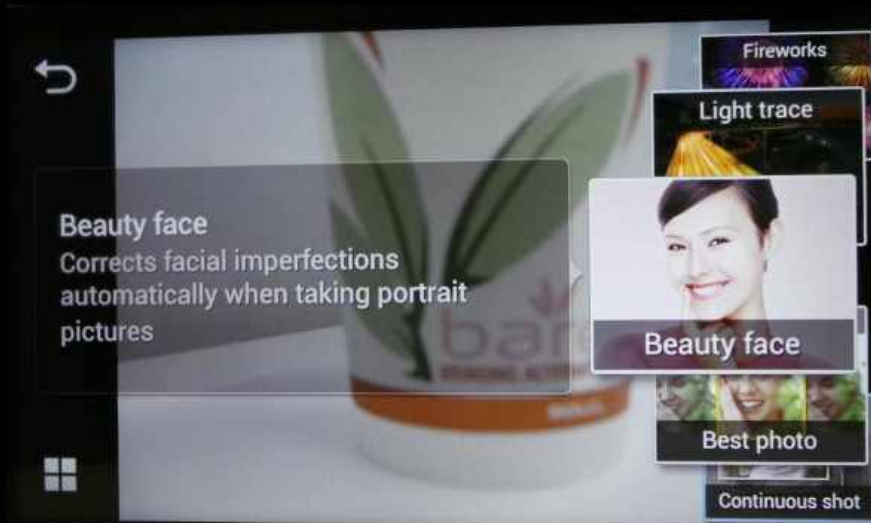
Galaxy Cameras exceed pocketability threshold.

Galaxy Camera Capture Fitness *(Camera UX)*

Finding Macro

[Watch video](#)

Galaxy Camera Capture Fitness *(Camera UX)*



Ferris Mode Wheel

It's a hide-and-seek slow wheel!
Does it take pictures with Macro?

Mode wheel with description is
more effective as help tip.

Eyes and hands work faster with the grid. All modes are visible. It should be the default Out-of-Box experience! Although press and hold "Mode" will get to the grid directly, it is difficult to discover. User education will be needed for this useful hidden feature.



Matrix Mode Grid

Capture the moment! Make scene modes quick!

Galaxy Camera Capture Fitness *(Camera UX)*

Capturing the
Precious Moment

[Watch video](#)

Galaxy Camera Capture Fitness *(Camera UX)*



Location of Home Icon, use of Back Icon at the same location and lack of space for physical grip (narrow bezel) with left hand make it more error prone .

My magic thumb touch again!
Good gracious! I did not lose precious moment this time...
I wish I can have a resizable viewfinder so that a wider bezel can be used for my magic thumbs.



Accidental Exit from camera can be minimized by design.

Galaxy Camera Performance

Good low light performance

**clipped from a Samsung
Swiss-Gear Backpack
Photo (uncompressed)**



Useful shooting modes



f/4.5 1/8s ISO800 31mm
(35mm equivalent: 172mm)



f/2.8 5s ISO100 4mm
(35mm equivalent: 23mm)

Great Picture Quality with many special effects and tools.

Galaxy Camera Performance

Hunting for focus in video

Sudden loss and hunting of focus happened infrequently especially with high zoom. The scene would also be over-exposed if auto had been used. Manual adjustments to the camcorder were made (as large as -1.7 EV in this clip).

Outstanding zoom: video and photo



f/4.5 1/40s ISO250 31mm
(35mm equivalent: 172mm)
Exposure bias -1.7

Manual adjustments to the camera or camcorder required for better picture. (as large as -1.7 EV in this photo).

Great zoom but with some hiccups in auto focus & exposure.

Galaxy Camera Photography Fitness *(Camera UX)*



This is a great Android camera!
Some suggestions of useful
photography apps and
bookmarks of photography
sites will be perfect.

Popular photography-related sites
can be provided through
bookmarks. Recommendations of
photography apps can be done by
S-Suggest widget or shortcut.

Where's my Photography Home?



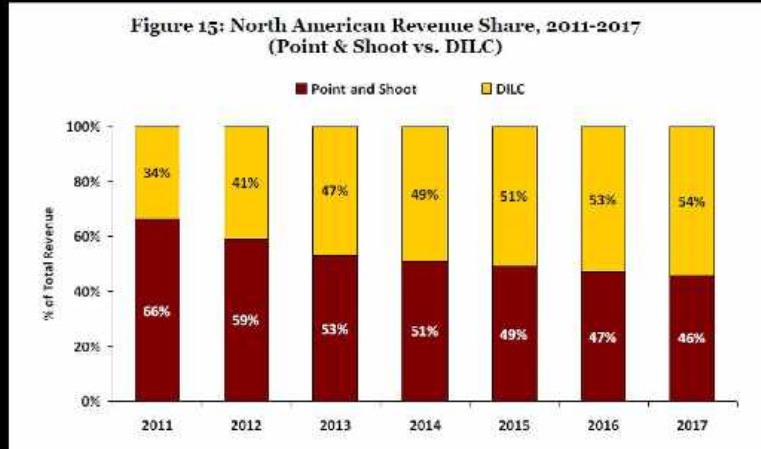
Example for
illustrative
purpose only

Need: Suggestions of Photography apps & resources for headstart

Competitive Thoughts – Product Category

Focus on Android
Compact
System Camera
with Interchangeable Lens?

Competitive Thoughts – Product Category



* 2012 North America Consumer Digital Camera Forecast: 2011-2017

Low-end DSCs are replaced by smartphone cameras. Demands of inexpensive digital cameras are projected to decline.



Samsung experience in Android and Galaxy brand equity will help to create a key differentiator from Nikon, breaking into this growing segment. (Sony will more likely to be our major competitor because of their smartphone experience)

One growth area is digital interchangeable lens cameras (DILC) including Digital SLRs and Compact System Interchangeable Cameras.



Sony NEX and Nikon 1 series as competitive benchmark.

Competitive Thoughts – physical form factor



Approx. 5.1 x 2.75 x 1.1" (WxHxD)



Approx. 4.75 x 2.75 x 1.1" (WxHxD)

The form factor of Galaxy Camera is bigger than many of the competitive compact system cameras with interchangeable lens. Although it is generally smaller than bridge camera or entry DSLR, performance and image quality is not as good.



The 4.8" screen is gorgeous but the sheer size of the display exacerbates the perceived risk of breaking the screen easily.

Need: Lean Form Factor to compete with Compact Interchangeable.

Competitive Thoughts – Camera UX



Focus on improvement of “automatic” capture setting intelligence for the best picture under varied environments. Accidental interaction with the huge touchscreen can be reduced by providing a resizable viewfinder with adjustable bezel width.

While the camera UX of Galaxy Camera is far superior to Nikon S800c, optimization of the capability to take good pictures and videos in more extreme situations will be needed. Auto focus capability & shutter lag are major areas in need of attention.



Need: Intelligent Camera – optimize usability and performance.

Competitive Thoughts – Photography Apps

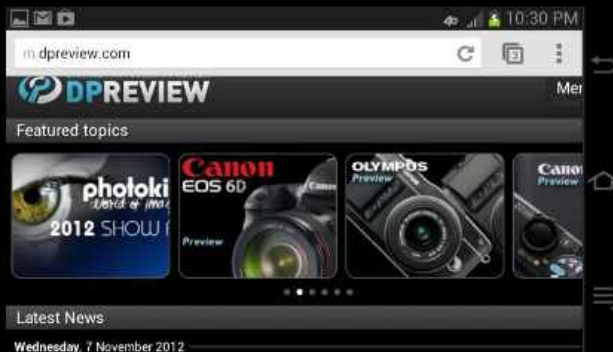
Photography resources at
fingertips in the camera.

Competitive Thoughts – Photography Apps



Photography oriented Home Pages with S-Suggest photography recommendations, tutorials, photography-related bookmarks and resources accessible any time, anywhere.

Web Resources (e.g., bookmarks) on Digital Cameras and Photography.



Need: Well connected to photography apps & resources @Home